

MODULE 39 · DOMAIN 4 - ACCELERATE WORKLOAD MIGRATION AND MODERNIZATION

Cutover and Validation

Written for people who will read it tired, at three in the morning.

DOMAIN 4 of 4 EXAM WEIGHT 20% MODULE 39 of 44

Every decision you have not made in advance, you will make at 3am, tired, under pressure. **Decide first.**

What a cutover runbook contains

- ◆ **Every activity, in sequence, with planned start and end times and a named owner.** A person, not a team.
- ◆ **The outage window,** and which tasks can run in parallel to compress it.
- ◆ **A rollback plan for exceeding the window,** not only for failing. Running late is the more common problem.
- ◆ AWS advises **at least two walkthrough meetings** with every task owner, including integration testers who sign off.

ON THE EXAM

"The networking team updates DNS" is not an owner. At 3am you need a name and a phone number, and that person needs to expect the call.

All at once, or phased?

ALL AT ONCE

One switch, everybody moves

- Flip DNS or the load balancer
- Necessary when both cannot run
- Hostname conflicts, licences, domain auth
- Rollback is another full switch

PHASED

A percentage at a time

- On-premises capacity stays available
- Rollback is a routing change
- Load increases gradually, so you can watch
- Both environments run, and cost, together

THE TIEBREAKER

AWS names three things that force all-at-once: hostname conflicts, licensing, and domain authentication. Look for those specifically.

The DNS problem

- ◆ A record with a 24-hour TTL takes up to 24 hours to move traffic, and the same to move it back.
- ◆ Lower TTLs to 60 or 300 seconds days before cutover, not on the night. The change itself needs the old TTL to expire.
- ◆ Raise them again once the migration is stable, since short TTLs mean far more queries.
- ◆ Some resolvers ignore TTLs. Plan for stragglers, and keep the old environment answering for a while.

ON THE EXAM

This is the single most common cutover mistake, and it is entirely preventable a week earlier. Module eight's caching problem, at the worst moment.

Rollback, decided in advance

- ◆ **Define checkpoints with criteria:** at this time, if this is not true, we roll back. Written down beforehand.
- ◆ **Name who decides.** A rollback argued about by six people at 4am is a rollback that happens too late.
- ◆ **A dual-write strategy** keeps both systems current, so reverting does not lose the transactions taken since cutover.
- ◆ **The point of no return** is a real moment. Know when it passes, and make sure everyone present knows too.

ON THE EXAM

The hardest part of a rollback is admitting it is needed. Predefined criteria remove that judgement from the tired people in the room.

Validation, and hypercare

- ◆ **Script the checks.** DNS answer, service ports, one real transaction. Nobody validates well from memory at 3am.
- ◆ **Validate functionally, then on performance, then on data.** Working is not the same as fast, and neither is the same as correct.
- ◆ **Hypercare** is a defined period of heightened support afterwards, with the migration team still engaged rather than dispersed.
- ◆ The application owner signs off, not the migration team. They know what correct looks like.

ON THE EXAM

A scripted check that runs in two minutes gets run. A checklist of forty manual steps gets skipped at the end of a long night.

Decommissioning, later than you think

- ◆ **Keep the source through at least one full business cycle.** Month-end and quarter-end jobs surface weeks after cutover.
- ◆ **Power off, observe, back up, then delete.** In that order, with a defined gap between each step.
- ◆ Exit hypercare before decommissioning begins. They are separate decisions, taken at separate times.
- ◆ Destroying the rollback in week one is the classic error. It converts a recoverable problem into an incident.

ON THE EXAM

A migration that went perfectly for three weeks can still fail at month end, on a job nobody remembered existed.

Reading the scenario

THE SCENARIO SAYS

Traffic did not move for hours after cutover

Hostname conflict or licence bound to a host

Gradual load increase and easy rollback

Rollback was debated during the incident

Month-end job failed after decommissioning

The runbook says "the DBA team"

IT IS POINTING AT

DNS TTL was not lowered in advance

All-at-once cutover is forced

Phased cutover

Predefined checkpoints and a named decider

Source kept for less than a business cycle

Name individuals, per AWS guidance

ON THE EXAM

Ask what was decided in advance. Almost every cutover failure traces back to a judgement left until the night itself.

Meridian Freight cut over its booking site at 02:00 by updating a Route 53 record. At 09:00 the on-premises servers were still receiving roughly 60% of traffic, and the two environments had diverged. The record's TTL was 86,400 seconds.

The next cutover is in three weeks.

What should the solutions architect recommend?

- ☐ A Lower TTLs to 60 seconds at least a day before the next cutover, and keep the old environment answering during transition
- ☐ B Lower the TTL to 60 seconds immediately before the next cutover
- ☐ C Use a shorter maintenance window so divergence has less time to occur
- ☐ D Shut down the on-premises servers at cutover so traffic has nowhere to go

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

- A** Lower TTLs a day ahead
The lowering itself only takes effect once the old 86,400-second value expires. Doing it a day early means the new value is live at cutover.
- B** Lower it immediately before
The sharpest distractor. Resolvers caching the old record still hold 86,400 seconds, so the change is invisible for up to a day.
- C** Shorter window
The maintenance window was not the problem. Traffic kept arriving for seven hours because of caching, not because the window was long.
- D** Shut down on-premises
Turns a divergence problem into an outage for 60% of users, and destroys the rollback at the same moment.

THE SIGNAL

Option B is the trap: lowering a TTL is itself subject to the old TTL. That recursion is what makes advance planning mandatory.

During a cutover, response times were three times normal. The team spent 90 minutes debating whether to roll back, with the migration lead arguing to continue and the application owner arguing to revert. By the time they decided, the point of no return had passed.

No rollback criteria had been written down.

What should the solutions architect recommend for future cutovers?

- ☐ A Define rollback checkpoints with measurable criteria and a single named decision-maker, agreed before cutover night
- ☐ B Give the migration lead sole authority to continue
- ☐ C Extend the maintenance window so there is more time to debate
- ☐ D Roll back automatically whenever response times exceed normal

◆ PAUSE AND DECIDE

What should the solutions architect recommend for future cutovers?

- A** Checkpoints, criteria, named decider
Moves the judgement to a calm moment. The debate happens in a planning meeting, and cutover night executes a decision already taken.
- B** Migration lead decides
Fixes the deadlock and picks the person with the strongest incentive to continue. Authority without criteria is still a judgement at 4am.
- C** Longer window
More time to have the same argument. The debate was not short of time; it was short of an agreed threshold.
- D** Automatic rollback on slowness
Overcorrects. Some slowness is expected as caches warm, so an automatic trigger would revert healthy cutovers.

THE SIGNAL

The failure is not that they chose wrongly. It is that a judgement requiring calm analysis was scheduled for 4am.

An application's licence is bound to a specific hostname, and the vendor will not issue a second licence for a parallel environment. The application also uses a fixed NetBIOS name that cannot exist twice in the domain.

The team's standard practice is a phased cutover with gradual traffic shifting.

What should the solutions architect recommend?

- ☐ A An all-at-once cutover, since the environments cannot run in parallel
- ☐ B A phased cutover as usual, using weighted DNS records
- ☐ C Rename the migrated application to avoid the conflict
- ☐ D Negotiate a second licence with the vendor

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

- A** All-at-once
AWS names hostname conflicts, licensing and domain authentication as exactly the constraints that force this approach. Two of the three are present.
- B** Phased as usual
Phased requires both environments running. The licence and the NetBIOS name each independently prevent that.
- C** Rename it
The licence is bound to the hostname, so renaming invalidates the licence. It solves one constraint by breaking the other.
- D** Second licence
The scenario says the vendor will not issue one. Worth asking, and not something to plan the cutover around.

THE SIGNAL

Phased cutover is better in almost every way and requires both environments to coexist. When they cannot, the choice is made for you.

A cutover completed successfully. Three weeks later, the month-end reconciliation job failed, because it depended on a file share on a decommissioned server. The source environment had been deleted in week one to save cost.

Recovery took four days and required restoring from an off-site backup.

What should the solutions architect recommend for future cutovers?

- A Retain the source through at least one full business cycle, then power off, observe, back up, and only then delete
- B Improve the dependency mapping so file shares are not missed
- C Take more frequent backups of the source before decommissioning
- D Delete the source only after the application owner signs off at cutover

◆ PAUSE AND DECIDE

What should the solutions architect recommend for future cutovers?

- A

 One full business cycle, staged deletion
Month-end and quarter-end jobs are exactly what a three-week retention misses. The staged sequence keeps recovery cheap throughout.
- B

 Better dependency mapping
Worth doing and it would not have caught this. A job that runs monthly makes no network connections during the discovery window.
- C

 More backups
They had a backup; recovery still took four days. The problem is time to recover, not whether a copy existed.
- D

 Sign-off at cutover
The owner did sign off, and correctly, because everything worked. The fault appeared three weeks later.

THE SIGNAL

Option B is the tempting one, and a monthly job is invisible to discovery that ran for a fortnight. Retention covers what mapping cannot.

A cutover runbook lists 60 tasks. Owners are recorded as "Network team", "DBA team" and "App team". On the night, three tasks stalled because nobody on shift knew they were responsible, and one owner was on annual leave.

The runbook had been circulated by email but never walked through.

What should the solutions architect recommend?

- ☐ A Name individual people per task, and hold at least two walkthrough meetings with every task owner present
- ☐ B Add more detail to each task description
- ☐ C Assign a single cutover manager to perform all 60 tasks
- ☐ D Automate all 60 tasks so no owners are needed

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

- A** Name people, hold walkthroughs
Exactly AWS's guidance. A named person can be checked for availability, and a walkthrough is where the leave clash surfaces beforehand.
- B** More task detail
The tasks were understood. Nobody knew they owned them, which is an assignment problem rather than a documentation one.
- C** One person does everything
Sixty tasks across networking, databases and applications, performed by one person, in one window. It removes parallelism and creates a bottleneck.
- D** Automate everything
Desirable over time and not achievable before the next cutover. Some tasks are approvals and validations by design.

THE SIGNAL

Both failures came from naming teams rather than people. A team cannot be on annual leave, which is exactly why the gap went unnoticed.

A retailer is cutting over an order platform and wants to increase load on AWS gradually while watching performance, reverting quickly if anything degrades. Both environments can run simultaneously, and the database is being kept in sync by DMS with change data capture.

The business is nervous after a previous failed cutover.

What should the solutions architect recommend?

- ☐ A A phased cutover shifting a growing percentage of traffic, with rollback by reverting the routing weights
- ☐ B An all-at-once cutover during a maintenance window
- ☐ C Run both environments indefinitely with no cutover
- ☐ D Cut over during the quietest hour to reduce exposure

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

- A** Phased, with weighted routing
Both environments can run, so the constraint that forced all-at-once in scenario three is absent. Rollback becomes a routing change.
- B** All-at-once
Available, and it discards the gradual load increase and the fast rollback the business specifically asked for.
- C** Never cut over
Pays for both environments forever and leaves the data centre exit unresolved. Phased is a transition, not a destination.
- D** Quietest hour
Sensible scheduling, and it is still all-at-once. A quiet hour reduces how many people notice, not the blast radius.

THE SIGNAL

Pair this with scenario three. Same choice, opposite answer, because here nothing prevents the two environments coexisting.

What to carry out of this module

- ◆ **Decide in advance what you will not want to decide at 3am.** Success criteria, rollback triggers, and who calls it.
- ◆ **Lower DNS TTLs days beforehand.** Lowering a TTL is itself subject to the old TTL, which is why it cannot wait.
- ◆ **Name people, not teams,** and walk the runbook through at least twice with every owner present.
- ◆ **Keep the source through a full business cycle.** Month-end jobs are invisible to dependency mapping.

ON THE EXAM

Phased cutover is better wherever it is possible. Hostname conflicts, licensing and domain authentication are what make it impossible.